FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Advanced), 2015

ANSWERS, HINTS & SOLUTIONS

HALF COURSE TEST –VI
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	A	C
2.	A	D	B
3.	C	B	D
4.	B	A	B
5.	C	A	B
6.	C	C	B
7.	D	B	D
8.	B	A	A
9.	D	D	A
10.	A	A	B
11.	D	A	C
12.	A	A	A
13.	D	B	D
14.	B	B	B
15.	A	D	C
16.	A	A	A
17.	A	A	B
18.	C	A	C
19.	C	B	D
1.	(A) → (p), (q), (r), (s) (B) → (p), (q), (r), (s) (C) → (p), (q) (D) → (r), (s)	(A) → (s) (B) → (p, r) (C) → (p, q) (D) → (p)	(A) → (q) (B) → (r) (C) → (p) (D) → (r)
2.	(A) → (p), (s) (B) → (p), (r) (C) → (q) (D) → (p)	(A) → (s) (B) → (p, q) (C) → (p, r, s) (D) → (r, s)	(A) → (p, r) (B) → (q, r) (C) → (r, s) (D) → (q, r, s)
3.	(A) → (q) (B) → (p), (q), (r), (s) (C) → (q) (D) → (p), (q), (r), (s)	(A) → (s) (B) → (r) (C) → (q) (D) → (p)	(A) → (r) (B) → (p, q) (C) → (q, r) (D) → (p, s)

Physics

PART – I

SECTION – A

1. $< a > = \frac{|\vec{v}_2 - \vec{v}_1|}{\Delta t} = \frac{\sqrt{3}v}{(2\ell / v)} = \frac{\sqrt{3}v^2}{2\ell}$

2. $2g - T = 2a$ …(i)
   $T - g = a$ …(ii)
   $T = kx$ …(iii)
   $x = 1.33$ cm

3. $v^2 = u^2 + 2as$
   $36 = 0 + 2 \times 2 \times S$
   $S = 9$ m

4. Velocity after 1 sec = $5\sqrt{2}$
   Radius of curvature = $\frac{(5\sqrt{2})^2}{5/\sqrt{2}} = 10\sqrt{2} = 14$ m

Chemistry

PART – II

SECTION – A

9. The compounds given in options (A) & (C) has no electron deficiency. The compound (B) has both electrophilic and nucleophilic site. While the compound (D) has got only electrophilic site.

10. Lother Meyer Curve is plotted between atomic volume and atomic mass. More is the atomic volume, higher is the position occupied.

11. Decrease in entropy during formation of four membered ring is higher than that of three membered ring. That's why formation of four membered ring is thermodynamically less feasible.

12. $K_w = \left[H^+\right]\left[OH^-\right]$

For neutral solution $\left[H^+\right]^2 = K_w \left(\text{since} \left[H^+\right] = \left[OH^-\right]\right)$

$-2\log\left[H^+\right] = -\log K_w = pK_w$

$pH = \frac{pKw}{2}$

13. $Mg(HCO_3)_2 + Ca(OH)_2 \rightarrow CaCO_3 + 2H_2O + MgCO_3$

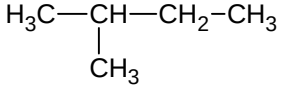
100ppm $\Rightarrow$ 10^6 g of water has 100g of $CaCO_3$

10 l of $H_2O \equiv 10^4$ g of water has $\frac{100}{10^6} \times 10^4$ g of $CaCO_3$ = 1g of $CaCO_3$

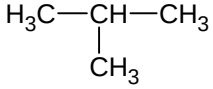
Moles of $CaCO_3 = \frac{1}{100}$ = moles of $Ca(OH)_2$

$\Rightarrow$ Mass of $Ca(OH)_2 = \frac{1}{100} \times 74 = 0.74$g

14. At high temperature, there is allyl substitution.

15. 

It has four types of replaceable H – atoms hence monochloro derivatives will be four. The product ratio will be 6 : 3 : 7.6 : 5. It is total number of hydrogen atoms multiplied by relative reactivity.

16. 

$1^\circ = 9 \times 1 = 9$

$3^\circ = 1 \times 1600 = 1600$

Thus the product ratio is 9 : 1600 and hence

% of $1^\circ = \frac{9}{1609} \times 100 \approx 0.5\%$

% of $3^\circ = \frac{1600}{1609} \times 100 \approx 99.5\%$

17. $\Delta x = r$

$\Delta x.\Delta p = \frac{h}{4\pi} \Rightarrow \Delta p = \frac{h}{4\pi.\Delta x} = \frac{h}{4\pi r}$

$\Delta p = p \Rightarrow p = \dfrac{h}{4\pi r}$

18. $\Delta x \times m\Delta V \geq \dfrac{h}{4\pi}$ or $\Delta V \geq \dfrac{6.626 \times 10^{-34}}{4 \times 3.14 \times 0.1 \times 10^{-10}}$ m sec^{-1}

$\geq 5.27 \times 10^{-24}$ m sec^{-1}

19. $\Delta V.\Delta x = \dfrac{h}{4\pi m} = \dfrac{6.626 \times 10^{-34}}{4 \times 3.14 \times 9.11 \times 10^{-31}} = 10^{-4}$ m^{2}s^{-1}

$\therefore \Delta V = \dfrac{10^{-4}}{10^{-8}} = 10^4$ ms^{-1}

Mathematics

PART – III

SECTION – A

1. Let the 1025th term falls in the nth group.
Then, $1 + 2 + 4 + ... + 2^{n-1} < 1025 \le 1 + 2 + 4 + ... + 2^n$
$\Rightarrow 2^{n-1} < 1026 \le 2^{n+1}$
$\Rightarrow n = 10$
$\Rightarrow 1025^{th}$ terms is 2^{10}

2. Here, ${}^{2n+1}C_1 + {}^{2n+1}C_2 + ... + {}^{2n+1}C_n = 63$
$\Rightarrow n = 3$

3. $|\omega z - 1 - \omega^2| = a$
$\Rightarrow |z + 1| = a \Rightarrow |z - 1 + 2| = a$
$\Rightarrow |z - 1| + 2 \ge a \Rightarrow 0 \le a \le 4$

4. $AG = \frac{1}{3}\sqrt{2b^2 + 2c^2 - a^2}$; $BG = \frac{1}{3}\sqrt{2a^2 + 2c^2 - b^2}$

$\Rightarrow AG = \frac{a}{3}$; $BG = \frac{1}{3}\sqrt{b^2 + 4c^2}$ as $a^2 = b^2 + c^2$

$\Rightarrow AG = \frac{5}{3}$; $BG = \frac{1}{3}\sqrt{16 + 36} = \frac{2}{3}\sqrt{13}$

Also, $AB = c = 3$ and $\Delta_{GAB} = \frac{1}{3}\Delta_{ABC} = 2$

If R_1 is the circumradius of triangle GAB, then

$R_1 = \frac{(AG)(BG)(AB)}{4\Delta_{GAB}} = \frac{5}{3} \times \frac{2}{3}\sqrt{13} \times 3 \times \frac{1}{4 \times 2} = \frac{5\sqrt{13}}{12}$ units

5. $\cos^{-1}(2x^2 - 1) = 2\pi - 2\cos^{-1} x$ (as $x < 0$)
$\Rightarrow \cos^{-1}(2x^2 - 1) - 2\sin^{-1} x = 2\pi - 2\cos^{-1} x - 2\sin^{-1} x$
$= 2\pi - 2(\cos^{-1} x + \sin^{-1} x) = 2\pi - 2\left(\frac{\pi}{2}\right) = \pi$

6. The line $2y = gx + \alpha$ should pass through $(-g, -g)$
So $-2g = -g^2 + \alpha \Rightarrow \alpha = g^2 - 2g = (g - 1)^2 - 1 \ge -1$

7. Tangents $y = m_1x + c$ and $y = m_2x + c$ intersect at $(0, c)$ which lies on the directrix of the given parabola
Hence, tangents are perpendicular for which $m_1m_2 = -1$

8. If (x_i, y_i) is the point of intersection of given curves

Then $\frac{\sum_{i=1}^{4} x_i}{4} = \frac{1+1}{2}$ and $\frac{\sum_{i=1}^{4} y_i}{4} = 0$

Now, $\frac{\sum_{i=1}^{3} x_i}{3} = \frac{4 - x_4}{3}$ and $\frac{\sum_{i=1}^{3} y_i}{3} = -\frac{y_4}{4}$

Centroid $\left(\frac{\sum_{i=1}^{3} x_i}{3}, \frac{\sum_{i=1}^{3} y_i}{3}\right)$ lies on the line $y = 3x - 4$

Hence, $-\frac{y_4}{3} = \frac{3(4 - x_4)}{3} - 4$

$\Rightarrow y_4 = 3x_4$

Hence, the locus of D is $y = 3x$

9. Clearly, $x^n - 1 = (x - 1)(x - a_1)(x - a_2) \ldots (x - a_{n-1})$

$\Rightarrow \frac{x^n - 1}{x - 1} = (x - a_1)(x - a_2) \ldots (x - a_{n-1})$

$\Rightarrow 1 + x + x^2 + \ldots + x^{n-1} = (x - a_1)(x - a_2) \ldots (x - a_{n-1})$

$\Rightarrow n = (1 - a_1)(1 - a_2) \ldots (1 - a_{n-1})$ (Putting $x = 1$)

10. $x = 1111 \ldots$ 91 times

$= 1 + 10 + 10^2 + \ldots + 10^{90} = \frac{10^{91} - 1}{10 - 1}$

$= \frac{10^{91} - 1}{10^{13} - 1} \times \frac{10^{13} - 1}{10 - 1}$

$= (1 + 10^{13} + 10^{26} + \ldots + 10^{78}) \times (1 + 10 + 10^2 + \ldots + 10^{12})$

= Composite number

11. $x^3 + x^2 + x = x(x^2 + x + 1) = x(x - \omega)(x - \omega^2)$

Now $f(x) = (x + 1)^n - x^n - 1$ is divisible by $x^3 + x^2 + x$, if $f(0) = 0$, $f(\omega) = 0$ and $f(\omega^2) = 0$

Now, $f(0) = (0 + 1)^n - 0^n - 1 = 0$

$f(\omega) = (\omega + 1)^n - \omega^n - 1 = (-\omega^2)^n - \omega^n - 1 = 0$

Similarly, $f(\omega^2) = 0$

12. Let $y = 27^{\cos 2x} \times 81^{\sin 2x} = 3^{3\cos 2x + 4\sin 2x}$

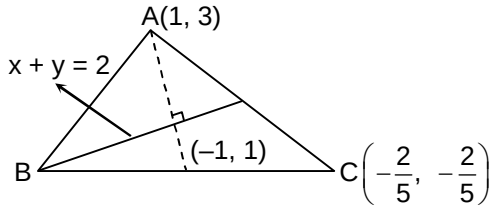
Now, $-\sqrt{3^2 + 4^2} \le 3\cos 2x + 4\sin 2x \le \sqrt{3^2 + 4^2}$

$\Rightarrow 3^{-5} \le 3^{3\cos 2x + 4\sin 2x} \le 3^5$

13. Locus of P is line segment between (–2, 0) and (2, 0)

14.-16. Image of A(1, 3) in line $x + y = 2$ is

$\left(1 - \frac{2(2)}{2},\ 3 - \frac{2(2)}{2}\right) \equiv (-1, 1)$



So line BC passes through (–1, 1) and $\left(-\frac{2}{5},\ -\frac{2}{5}\right)$

The equation of line BC is $y - 1 = \frac{\left(-\frac{2}{5} - 1\right)}{\left(-\frac{2}{5} + 1\right)}(x + 1)$

$\Rightarrow 7x + 3y + 4 = 0$

Vertex B is point of intersection of $7x + 3y + 4 = 0$ and $x + y = 2$, i.e., $B \equiv \left(-\frac{5}{2},\ \frac{9}{2}\right)$

Line AB is $y - 3 = \frac{\left(3 - \frac{9}{2}\right)}{\left(1 + \frac{5}{2}\right)}(x - 1) \Rightarrow 3x + 7y = 24$

17.-19. If no box remains empty, we can have (1, 1, 3) or (1, 2, 2) distribution pattern

17. Here, number of ways $= \frac{5!}{1!\ 2!\ 3!} + \frac{5!}{(2!)^2\ 1!\ 2!} = 25$

18. Here, we have only two ways (1, 1, 3) and (1, 2, 2)
Hence, number of ways is 2

19. When boxes are kept in a row, they will be treated as different. In this case, the number of ways will be ${}^{5-1}C_{3-1} = {}^4C_2 = 6$

SECTION – B

1. (A) a, b, c are in G.P. Hence, $b^2 = ac$

$\Rightarrow 2\log_{10} b = \log_{10} a + \log_{10} c$

$\Rightarrow \dfrac{2}{\log_b 10} = \dfrac{1}{\log_a 10} + \dfrac{1}{\log_c 10}$

$\Rightarrow \dfrac{2}{y} = \dfrac{1}{x} + \dfrac{1}{z}$

Hence, x, y, z are in H.P.

(B) $\dfrac{a+be^x}{a-be^x} = \dfrac{b+ce^x}{b-ce^x} = \dfrac{c+de^x}{c-de^x}$

$\Rightarrow \dfrac{2a}{a-be^x} - 1 = \dfrac{2b}{b-ce^x} - 1 = \dfrac{2c}{c-de^x} - 1$

$\Rightarrow \dfrac{a-be^x}{a} = \dfrac{b-ce^x}{b} = \dfrac{c-de^x}{c}$

$\Rightarrow$ a, b, c, d are in G.P.

(C) Given, $2b = a + c$, $x^2 = ab$, $y^2 = bc$

Now, $x^2 + y^2 = b(a + c) = 2b^2$

$\Rightarrow x^2 + y^2 = 2b^2$

Hence, x^2, b^2, y^2 are in A.P.

(D) $x \log a = y \log b = z \log c = k$ (say)

Also, $y^2 = xz$

$\Rightarrow \dfrac{k^2}{(\log b)^2} = \dfrac{k^2}{\log a \log c}$

$\Rightarrow$ log a, log b, log c are in G.P.

2. (A) $\cot\dfrac{A}{2} = \dfrac{b+c}{2}$

$\Rightarrow \dfrac{\cos\left(\frac{A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \dfrac{\sin B + \sin C}{\sin A} = \dfrac{2\sin\frac{B+C}{2}\cdot\cos\frac{B-C}{2}}{2\sin\frac{A}{2}\cos\frac{A}{2}}$

$\Rightarrow \cos\dfrac{A}{2} = \cos\dfrac{B-C}{2}$

$A = B - C \Rightarrow A + B + C = 2B = \pi \Rightarrow B = \dfrac{\pi}{2}$

(B) In ΔABC, $a\tan A + b\tan B = (a + b)\ \tan\left(\dfrac{A+B}{2}\right)$

$\Rightarrow a\left[\tan A - \tan\left(\dfrac{A+B}{2}\right)\right] = b\left[\tan\left(\dfrac{A+B}{2}\right) - \tan B\right]$

$\dfrac{a\sin\left(A - \frac{A+B}{2}\right)}{\cos A\cos\left(\frac{A+B}{2}\right)} = \dfrac{b\sin\left(\frac{A+B}{2} - B\right)}{\cos\left(\frac{A+B}{2}\right)\cos B}$

$\Rightarrow \dfrac{2R\sin A\sin\left(\dfrac{A-B}{2}\right)}{\cos A} = \dfrac{2R\sin B\sin\left(\dfrac{A-B}{2}\right)}{\cos B}$

$\Rightarrow \sin\left(\dfrac{A-B}{2}\right)(\tan A - \tan B) = 0$

$\Rightarrow A = B$

(C) $a\cos A = b\cos B$

$2R\sin A\cos A = 2R\sin B\cos B$

$\Rightarrow \sin 2A = \sin 2B$

$\Rightarrow 2A = 2B$ or $2A = \pi - 2B$

$\Rightarrow A = B$ or $A + B = \dfrac{\pi}{2}$

(D) $\cos A = \dfrac{\sin B}{2\sin C}$

$\Rightarrow \dfrac{b^2 + c^2 - a^2}{2bc} = \dfrac{b}{2c}$

$\Rightarrow b^2 + c^2 - a^2 = b^2$

$\Rightarrow c^2 = a^2 \Rightarrow c = a \Rightarrow \Delta$ is isosceles

3. (A) Radical axis of $x^2 + y^2 + 2a_1x + b = 0$ and $x^2 + y^2 + 2a_2x + b = 0$ is $(a_1 - a_2)x = 0$ or $x = 0$

It must touch both the circles

Solving it with one of the circles we get $y^2 + b = 0$

$\Rightarrow b \le 0$

(B) Radical axis of given circle is $a_1x - a_2y = 0$

Solving it with one of the circles, we have $x^2 + \left(\dfrac{a_1}{a_2}\right)x^2 + 2a_1x + b = 0$

This must have equal roots

Hence, $4a_1^2 - 4b\left[1 + \dfrac{a_1^2}{a_2^2}\right] = 0$

$\Rightarrow a_1^2 - b\left(1 + \dfrac{a_1^2}{a_2^2}\right) = 0$

Options (p) and (q) satisfy this

(C) If the straight line $a_1x - by + b^2 = 0$ touches the circle $x^2 + y^2 = a_2x + by$

$\Rightarrow \dfrac{\left|a_1 \cdot \dfrac{a_2}{2} - b \cdot \dfrac{b}{2} + b^2\right|}{\sqrt{a_1^2 + b^2}} = \sqrt{\dfrac{a_2^2}{4} + \dfrac{b^2}{4}}$

$\Rightarrow a_1^2a_2^2 + 2b^2a_1a_2 + b^4 = a_1^2a_2^2 + a_1^2b^2 + a_2^2b^2 + b^4$

$\Rightarrow b^2 = 0$ or $2a_1a_2 = a_1^2 + a_2^2$

$\Rightarrow$ Options (q) and (r) satisfy this

(D) Line $3x + 4y - 4 = 0$ touches the circle $(x - a_1)^2 + (y - a_2)^2 = b^2$

$\Rightarrow \dfrac{|3a_1 + 4a_2 - 4|}{5} = b$